Five Fundamental Constants from Two Integers: Dark Energy, the Fine Structure Constant, the Weinberg Angle, and the Proton-Electron Mass Ratio from $d = 4$ and $\mathrm{cd} = 3$

Wright Brothers[1]

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We observe that five dimensionless physical constants can be expressed, with no free parameters, in terms of two integers: the spacetime dimension $d = 4$ and the étale cohomological dimension $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ (Artin-Verdier), together with π. Defining $r = d \cdot \mathrm{cd} = 12$ and $r' = \mathrm{cd} \cdot (d+1) = 15$, which satisfy $r^3 = j(i) = 1728$ and $r'^3 = |j(\frac{1+i\sqrt{7}}{2})| = 3375$ (values of the modular j-invariant at CM points):

$$\begin{aligned}
\Omega_\Lambda &= 2\pi/\mathrm{cd}^2 &&= 0.698 \quad (2.0\%)\\
\alpha_{\mathrm{EM}} &= 4\pi/(d\mathrm{cd})^3 &&= 1/137.5 \quad (0.3\%)\\
\sin^2\theta_W &= \tfrac{\mathrm{cd}}{8\pi}\left(\tfrac{d+1}{d}\right)^3 &&= 0.233 \quad (0.8\%)\\
m_p/m_e &= d\,\mathrm{cd}^3(d^2+1) &&= 1836 \quad (0.008\%)\\
m_\mu/m_e &= \mathrm{cd}^2 \times 23 &&= 207 \quad (0.11\%)
\end{aligned}$$

The first four follow from d and cd alone. The consistency $j(i) = (d \cdot \mathrm{cd})^3$ is not assumed but verified: it connects the physics (dimension theory) to the mathematics (j-invariant at CM points). Coupling constants involve π (archimedean); mass ratios are purely arithmetic (no π). We discuss the structural origin of each formula, the risk of numerology, and experimental tests.

THE TWO INTEGERS

Let d denote the spacetime dimension and cd the étale cohomological dimension of $\mathrm{Spec}(\mathbb{Z})$. The former is conventionally $d = 3+1 = 4$; the latter is a theorem of Artin and Verdier [?]: $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$. Both are mathematically rigid: d counts large-scale dimensions, cd is a property of the integers proved in the 1960s.

Derived quantities

From $d = 4$ and $\mathrm{cd} = 3$:

$$r \equiv d \cdot \mathrm{cd} = 12, \tag{1}$$

$$r' \equiv \mathrm{cd} \cdot (d+1) = 15, \tag{2}$$

$$d^2 + 1 = 17 \quad \text{(Fermat prime)}, \tag{3}$$

$$r^3 = 1728 = j(i), \tag{4}$$

$$r'^3 = 3375 = |j(\tfrac{1+i\sqrt{7}}{2})|. \tag{5}$$

The identities (4) and (5) are *not assumed*. They are verifiable facts: $j(i) = 1728$ is a classical result in the theory of complex multiplication [?], and $j(\frac{1+i\sqrt{7}}{2}) = -3375$ follows from the class field theory of $\mathbb{Q}(\sqrt{-7})$ [?]. That r^3 and r'^3 equal these j-values is a *consistency check* on the identification $r = d \cdot \mathrm{cd}$, $r' = \mathrm{cd}(d+1)$.

FIVE PREDICTIONS

Dark energy: $\Omega_\Lambda = 2\pi/\mathrm{cd}^2$

In previous work [?], we derived the dark energy density parameter from the Bost-Connes partition function $\zeta(\beta)$ [?] and the Wheeler-DeWitt temporal boundary condition:

$$\Omega_\Lambda = \frac{d}{\mathrm{cd}}\,|\xi_{\neg 2}(-1)| = \frac{d}{\mathrm{cd}} \cdot \frac{\pi}{6} = \frac{2\pi}{9} = \frac{2\pi}{\mathrm{cd}^2}. \tag{6}$$

Numerically: 0.6981; observed (Planck 2018): 0.6847 ± 0.0073; discrepancy: 1.96% (1.8σ).

The simplification $\Omega_\Lambda = 2\pi/\mathrm{cd}^2$ uses $d/\mathrm{cd} = 4/3$ and $\pi/6 = \pi/(2\mathrm{cd})$, giving $2\pi/\mathrm{cd}^2$.

Fine structure constant: $\alpha = 4\pi/(d\mathrm{cd})^3$

$$\alpha_{\mathrm{EM}} = \frac{4\pi}{r^3} = \frac{4\pi}{(d\mathrm{cd})^3} = \frac{4\pi}{1728} = \frac{1}{137.51}. \tag{7}$$

Observed (CODATA 2018): $\alpha^{-1} = 137.036$; discrepancy: 0.34%.

Since $r^3 = j(i)$, this is equivalently $\alpha = 4\pi/j(i)$: the fine structure constant is the ratio of the "geometric circle" (4π) to the "arithmetic cube" ($j(i)$).

Weinberg angle: $\sin^2\theta_W = \frac{\mathrm{cd}}{8\pi}\left(\frac{d+1}{d}\right)^3$

$$\sin^2\theta_W = \frac{\mathrm{cd}}{8\pi}\left(\frac{d+1}{d}\right)^3 = \frac{3}{8\pi}\left(\frac{5}{4}\right)^3 = \frac{375}{512\pi} = 0.2331. \tag{8}$$

Observed (PDG 2024): 0.23121 ± 0.00004; discrepancy: 0.84%.

This decomposes as

$$\sin^2\theta_W = \underbrace{\frac{3}{8}}_{\text{SU(5) GUT}} \times \underbrace{\frac{1}{\pi}\left(\frac{d+1}{d}\right)^3}_{\text{running factor}}. \tag{9}$$

The GUT prediction 3/8 is exact at unification. The "running factor" $(d+1)^3/(d^3\pi)$ encodes the renormalization group flow from the GUT scale to m_Z, expressed purely in terms of the spacetime dimension.

The ratio $r'/r = (d+1)/d = 5/4$ equals the ratio of j-invariant cube roots at two CM points: $|j(-7)|^{1/3}/j(-4)^{1/3} = 15/12 = 5/4$. The Weinberg angle is the "angular distance" between the $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-7})$ sectors of the modular curve.

Proton-electron mass ratio: $m_p/m_e = d\cdot\mathrm{cd}^3(d^2+1)$

$$\frac{m_p}{m_e} = d\cdot\mathrm{cd}^3\cdot(d^2+1) = 4\times 27\times 17 = 1836. \tag{10}$$

Observed (CODATA 2018): 1836.15267343(11); discrepancy: 0.0083%.

This is the most precise prediction (40× better than α). It admits a Gaussian-norm interpretation: defining $z = r + \mathrm{cd}\,i = 12 + 3i \in \mathbb{Z}[i]$,

$$\frac{m_p}{m_e} = \mathrm{Re}(z)\times N_{\mathbb{Z}[i]}(z) = 12\times|12+3i|^2 = 12\times 153 = 1836, \tag{11}$$

where $N_{\mathbb{Z}[i]}(z) = z\bar{z} = 144 + 9 = 153$.

The Gaussian integer $z = 12 + 3i$ encodes:

- $\mathrm{Re}(z) = d\mathrm{cd} = 12$: the electromagnetic scale.
- $\mathrm{Im}(z) = \mathrm{cd} = 3$: the QCD scale (number of colors).
- $N(z) = \mathrm{cd}^2(d^2+1) = 9\times 17 = 153$: the combined magnitude.

The factor $d^2+1 = 17$ is a Fermat prime ($17 = 2^{2^2}+1$) and divides the order of the Monster group.

Muon-electron mass ratio: $m_\mu/m_e = \mathrm{cd}^2\times 23$

$$\frac{m_\mu}{m_e} = \mathrm{cd}^2\times 23 = 9\times 23 = 207. \tag{12}$$

Observed: 206.768; discrepancy: 0.11%.

The prime 23 divides $|\mathbb{M}|$ (Monster group) and equals $24 - 1 = c(V^\natural) - 1$, where $c = 24$ is the central charge of the Moonshine module. A structural derivation of 23 from d and cd alone remains open.

ARITHMETIC-GEOMETRIC SEPARATION

Observation 1. *Coupling constants involve π; mass ratios do not.*

Constant	Formula	Contains π?
Ω_Λ	$2\pi/\mathrm{cd}^2$	Yes
α_{EM}	$4\pi/(d\mathrm{cd})^3$	Yes
$\sin^2\theta_W$	$\frac{\mathrm{cd}}{8\pi}(\frac{d+1}{d})^3$	Yes
m_p/m_e	$d\,\mathrm{cd}^3(d^2+1)$	**No**
m_μ/m_e	$\mathrm{cd}^2\times 23$	**No**

This separation is natural from the adelic viewpoint: $\mathbb{A}_\mathbb{Q} = \mathbb{R}\times\prod_p\mathbb{Q}_p$. Couplings involve the archimedean place (π from $\mathbb{R}$); mass ratios are purely non-archimedean (integer arithmetic). Physically: couplings measure *phase accumulation* (geometric), while masses measure *norms in number fields* (algebraic).

THE j-INVARIANT CONSISTENCY CHECK

The identities $r^3 = j(i)$ and $r'^3 = |j(-7)|$ are *verifiable predictions*, not inputs.

$j(i) = 1728$ is determined by the theory of CM elliptic curves over $\mathbb{Q}(i)$ [**?**]. The assertion $r = d\mathrm{cd} = 12$, hence $r^3 = 1728$, is determined by dimension theory (GR + Artin-Verdier). These two facts are *independent*: one is number theory, the other is physics. Their agreement is evidence for the framework's self-consistency.

Similarly, $|j(\frac{1+i\sqrt{7}}{2})| = 3375 = 15^3$ is number theory, while $r' = \mathrm{cd}(d+1) = 15$ is physics. Their agreement provides a second consistency check.

Conjecture 2 (Arithmetic-physical correspondence)**.** *The j-invariant at CM points of class-number-one imaginary quadratic fields takes the form $j =$ (linear combination of d and cd)^3, and these cube roots determine the coupling constants and mass ratios of the Standard Model.*

NUMEROLOGY ASSESSMENT

Against numerology. (1) Two parameters (d, cd) predict five quantities, leaving three overconstrained degrees of freedom. The probability that three random predictions match observations to $< 2\%$ is $\lesssim (0.04)^3 \approx 6\times 10^{-5}$. (2) The m_p/m_e match at 0.008% is an integer

identity ($1836 = 4 \times 27 \times 17$), not a continuous approximation. (3) The j-value consistency checks ($1728 = 12^3$, $3375 = 15^3$) are binary (they either hold or not) and cannot be "tuned." (4) The arithmetic-geometric separation (π in couplings only) is a structural prediction, not a numerical one.

For numerology (honest caveats). (1) The formulas are *observations*, not derivations from a Lagrangian or action principle. (2) The coefficient 4π in α (rather than 2π or π) lacks a first-principles explanation. (3) The muon formula requires 23, which is not expressed in terms of d and cd alone. (4) The 0.15 residual in m_p/m_e (1836 vs. 1836.153) is unexplained.

EXPERIMENTAL FALSIFIABILITY

The framework makes sharp predictions:

1. $\Omega_\Lambda = 0.6981$, testable by Euclid (2028, projected 0.5% precision).

2. $\alpha_{\mathrm{EM}} = 4\pi/1728$ is already 0.34% off; improved measurements or theoretical corrections could confirm or rule this out.

3. The Euler product test (182:1 vs. 1:3 in a 2-prime-sieved BAW cavity, $2,200) [**?**] validates the arithmetic vacuum structure underlying all five formulas.

DISCUSSION

If the five formulas are not coincidental, they imply that the Standard Model's dimensionless parameters are determined by the *arithmetic geometry of* $\mathrm{Spec}(\mathbb{Z})$ through two data: the spacetime dimension d and the cohomological dimension cd.

The physical content of d and cd is well understood: $d = 4$ counts spacetime dimensions; $\mathrm{cd} = 3$ is the "arithmetic dimension" of the integers, reflecting the structure of primes via étale cohomology. The remarkable fact is that these two integers, combined with the archimedean constant π, suffice to determine the vacuum energy, the electromagnetic coupling, the electroweak mixing, and the dominant mass ratio of baryonic matter.

The framework suggests a division of physics into an *arithmetic layer* (masses, generations, gauge groups—determined by $\mathrm{Spec}(\mathbb{Z})$) and a *geometric layer* (couplings, phases, propagation—determined by $\mathbb{R}$ and π), unified through the adelic structure of $\mathbb{Q}$.

Computations with PARI/GP, SageMath, and mpmath.